

NarraLens

AI-Generated Data Report

9056ccc9-2d23-40a8-b607-49af54daa350

100 rows × 12 columns

Key Insights

■ Average Index

50.5

Median: 50.5

■ Anomalies Detected

5

Records flagged as unusual by
IsolationForest

■ Missing Values

0

Total missing entries across all
columns

★ Dataset Size

100

12 columns analyzed

Statistical Summary

Column	Mean	Median	Std Dev	Min	Max	Missing%
Index	50.5	50.5	29.012	1	100	0.0%

AI-Generated Narrative Report

Executive Summary

This dataset comprises a customer contact registry of 100 records spanning 12 attribute columns, with a remarkably clean data profile — zero missing values across all fields and no detected outliers in the sole numeric dimension. While the dataset is structurally sound and complete, 5 anomalies were flagged during automated detection, and the predominantly categorical nature of the data limits deep statistical inference without further temporal or transactional enrichment.

Dataset Overview

The dataset represents a **customer contact directory**, likely extracted from a CRM (Customer Relationship Management) system or marketing database. It contains **100 rows and 12 columns**, capturing a range of personally identifiable and organizational information for each customer record.

Column Inventory

Column	Type	Description
Index	Numeric	Sequential row identifier (1–100)
Customer Id	Categorical	Unique alphanumeric customer identifier
First Name	Categorical	Customer's given name
Last Name	Categorical	Customer's family name
Company	Categorical	Associated organization
City	Categorical	Customer's city of residence/operation
Country	Categorical	Customer's country
Phone 1	Categorical	Primary contact number
Phone 2	Categorical	Secondary contact number
Email	Categorical	Email address
Subscription Date	Categorical	Date of subscription (stored as string, not parsed as datetime)

| Website | Categorical | Associated web presence |

Data Quality Assessment

The data quality profile is **exceptionally strong** for a dataset of this type:

- Total missing values: 0 across all 100 rows and 12 columns — a 100% completeness rate
- No numeric outliers detected in the Index column
- 5 anomalies flagged by automated detection logic, suggesting minor structural or formatting irregularities in categorical fields
- A notable concern is that Subscription Date was not parsed as a datetime column, indicating potential inconsistency in date formatting that prevented automatic type inference — this is a data quality risk that warrants remediation
- Only 1 numeric column (Index) is present, which serves as a row counter rather than a meaningful analytical variable; all substantive fields are categorical

Key Findings

- **Perfect Data Completeness:** The dataset achieves a 100% fill rate with zero missing values across all 12 columns and 100 records. This is statistically rare in real-world CRM datasets and suggests either a well-governed data entry process or that the dataset has been pre-cleaned prior to delivery.
- **High Cardinality in Identity Fields:** The Customer Id column yields 100 unique values across 100 records — a perfect 1:1 uniqueness ratio, confirming no duplicate customer identifiers exist. This is the expected behavior for a primary key field and validates the integrity of the customer registry.
- **Moderate Name Duplication:** The First Name field contains 93 unique values out of 100, with 6 names appearing twice — specifically Sheryl, Joanna, Faith, Shane, Tom, Sherry, and Lynn (each with a count of 2, representing 2.0% frequency). Similarly, Last Name contains 97 unique values with Mata, Foley, and Greer each appearing twice. This is expected in any sufficiently sized contact list and does not indicate data quality issues.
- **Subscription Date Not Datetime-Parsed:** Despite being a clearly temporal field, Subscription Date was categorized as a string/categorical type rather than a datetime column — yielding 0 datetime columns in the parsed schema. This prevents any temporal trend analysis and represents a structural data type issue requiring correction.
- **5 Anomalies Detected Across Categorical Fields:** Automated anomaly detection flagged 5 records with unusual patterns. Given that the only numeric column has zero outliers, these anomalies are almost certainly rooted in the categorical columns — likely irregular formatting in phone numbers, email

addresses, Customer IDs, or website URLs.

- **Coefficient of Variation of 57.45% on Index:** The Index column exhibits a CV of 57.45%, which is mathematically consistent with a uniform sequential distribution from 1 to 100. The perfectly symmetric skewness of 0.0 and platykurtic kurtosis of -1.2 confirm a flat, uniform distribution — as expected for a simple row counter.

Statistical Deep Dive

Index Column Analysis

The `Index` column is the dataset's only numeric field and functions purely as a sequential row identifier. Its statistical properties are entirely consistent with a uniform integer distribution:

- Mean = Median = 50.5: Perfect central symmetry, indicating no gaps or duplications in the sequential numbering
- Standard Deviation: 29.01, with Q1 at 25.75 and Q3 at 75.25, yielding an IQR of 49.5
- Skewness: 0.0 — perfectly symmetric distribution with no tail bias
- Kurtosis: -1.2 — a platykurtic (flat) distribution, which is the theoretical signature of a uniform distribution, confirming rows are evenly distributed without clustering
- Range: 1 to 100 with no gaps and no outliers (lower bound: -48.5, upper bound: 149.5 via IQR method)
- 98% of values fall above the mean as reported in peak insights — this is a nuanced statistical artifact worth noting; in a symmetric distribution, approximately 50% of values should be above the mean, suggesting the `pct_above_mean` metric may be calculated using a strict inequality that includes the mean value itself

Categorical Field Distributions

Given the absence of additional numeric columns, variability analysis for categorical fields is limited to frequency distributions:

- Customer Id: 100 unique values / 100 records = 100% uniqueness — effectively a primary key
- First Name: 93 unique values, top frequency 2.0% — low repetition, consistent with a diverse customer base
- Last Name: 97 unique values, top frequency 2.0% — even lower repetition than first names, again expected

The high cardinality across all identity fields (names, IDs, emails, phone numbers, websites) is characteristic of contact-type datasets and limits the utility of traditional frequency-based categorical analysis.

Correlations & Relationships

Quantitative Correlations

The `top_correlations`` array returned **empty** (`[]``), which is a direct consequence of having only one numeric column (`Index``). Correlation analysis requires at least two numeric variables, making traditional Pearson or Spearman correlation calculations inapplicable to this dataset in its current form.

Implicit Structural Relationships

While mathematical correlations cannot be computed, several **logical relational structures** can be inferred:

- **Customer Id ↔ All Fields:** The Customer Id serves as a foreign key linkage point. Its 1:1 relationship with all other fields means each record is atomically unique, with no fan-out or fan-in relationships detectable at the surface level.
- **First Name + Last Name ↔ Identity:** The combination of first and last names yields near-complete uniqueness (93×97 possible combinations), though the marginal duplication in both fields raises a low-probability risk of full-name collisions that should be cross-validated against Customer IDs.
- **Company ↔ City/Country:** In a typical CRM dataset, company affiliation correlates with geographic location. This relationship cannot be quantified from the current summary but would be analytically valuable with full data access.
- **Subscription Date ↔ Engagement Metrics:** As a temporal marker, Subscription Date would ideally correlate with customer activity or tenure metrics — but without datetime parsing or transactional data, this relationship remains latent and unexplorable.

Recommendations for Enrichment

To enable meaningful correlation analysis, the dataset would benefit from the addition of numeric fields such as purchase count, revenue value, account age (derived from Subscription Date), or engagement scores.

Anomalies & Outliers

Numeric Outliers

The IQR-based outlier detection on the `Index` column returned **0 outliers (0.0%)**, with bounds set at:

- Lower bound: -48.5
- Upper bound: 149.5

Since Index values range only from 1 to 100, all values comfortably fall within these bounds. This result is expected and carries no analytical significance.

Categorical Anomalies

5 anomalies were detected through automated pattern analysis on categorical fields. While the specific records are not identified in the summary, the most likely sources based on field characteristics include:

- **Malformed Email Addresses:** Emails missing @ symbols, containing double dots, or using non-standard TLDs
- **Irregular Phone Number Formats:** Phone 1 or Phone 2 fields with inconsistent formatting — e.g., mixed use of dashes, parentheses, spaces, or international dialing codes
- **Non-Standard Customer IDs:** The top Customer ID (DD37Cf93aecA6Dc) uses a mixed-case alphanumeric format; anomalies may reflect IDs that deviate from the established length or character pattern
- **Website URL Anomalies:** Missing protocol prefixes (http:// or https://), malformed domain structures, or placeholder values
- **Subscription Date Format Inconsistencies:** Dates entered in multiple formats (e.g., MM/DD/YYYY vs. YYYY-MM-DD) — which likely also explains why the column failed datetime parsing

The **5% anomaly rate** (5/100 records) is within an acceptable threshold for a raw CRM export but should be resolved before any downstream analytics or marketing automation processes consume this data.

Trends & Patterns

Temporal Trends

No datetime columns were successfully parsed in this dataset, which **completely precludes time-series or temporal trend analysis**. The `Subscription Date` column — which would typically be the primary driver of subscription cohort analysis, churn modeling, or growth trend identification — remains locked in

string format.

Had this column been properly parsed, it would be possible to analyze:

- Subscription velocity: Rate of new customer acquisitions over time
- Seasonal patterns: Monthly or quarterly clustering of subscriptions
- Cohort behavior: Grouping customers by subscription period to compare engagement or retention

Structural Patterns

Despite the absence of temporal data, several **structural patterns** emerge:

- Geographic Distribution: The presence of City and Country columns suggests a geographically distributed customer base. Without aggregated frequency data for these fields, the precise distribution is unknown — but the high cardinality typical of such datasets implies customers span multiple countries and cities.
- Corporate vs. Individual Customers: The Company field indicates this may be a B2B or mixed B2B/B2C contact database. Analyzing company name frequency would reveal whether the customer base skews toward individual consumers or organizational accounts.
- Dual Phone Number Prevalence: The inclusion of both Phone 1 and Phone 2 for all 100 records (given zero missing values) suggests either a data entry policy requiring two contact numbers or that the source system automatically populated secondary phone fields — potentially with duplicate or placeholder values worth investigating.

Recommendations

1. Remediate Subscription Date Parsing Immediately

The `Subscription Date` column's failure to parse as a datetime type is the single highest-priority data quality issue in this dataset. Standardize all date entries to **ISO 8601 format (YYYY-MM-DD)**, re-ingest the data, and validate that all 100 dates parse correctly. This will unlock cohort analysis, subscription trend modeling, and customer tenure calculations — transforming this from a static contact list into a time-aware analytical asset.

2. Investigate and Resolve the 5 Flagged Anomalies

Conduct a row-level audit of the 5 anomalous records. Create a validation pipeline that enforces:

- Email: RFC 5322-compliant regex validation
- Phone: E.164 international format standardization

- Website: URL schema validation with protocol enforcement
- Customer ID: Length and character-class pattern matching

Document the remediation for each record and implement these validation rules as upstream data entry constraints to prevent recurrence.

3. Enrich the Dataset with Behavioral or Transactional Numeric Variables

In its current state, the dataset contains only one numeric column (a row index), making quantitative analysis — including correlations, regression modeling, and predictive analytics — impossible. Prioritize joining this contact dataset with transactional records (purchase history, login frequency, support tickets) or appending derived metrics (account age from Subscription Date, email engagement scores) to enable meaningful statistical modeling.

4. Implement a Geographic Segmentation Analysis

Leverage the `City` and `Country` columns to build